FRONT

hovid

hovid SEFEPIM FOR INJECTION 1g

VISEF01-reg

DESCRIPTION
Finished Product : White to pale yellow powder
After reconstitution : Clear to pale yellow solution

COMPOSITION
hovid Sefepim for Injection 1g: Each vial contains Cefepime Hydrochloride equivalent to Cefepime 1g.

ACTIONS AND PHARMACOLOGY
Cefepime is a fourth-generation cephalosporin and is active against a wide range of Gram-positive such as staphylococci (but not meticililin-resistant *Staphylococcus aureus*) and streptococci and Gram-negative aerobic organisms.

It has a broader spectrum of activity against Enterobacteriaceae, including activity against organisms producing chromosomally mediated beta-lactamases such as *Enterobacter spp.* and *Proteus vulgaris*.

It has similar or slightly less activity against *Pseudomonas aeruginosa* than ceftazidime, although it may be active against some strains resistant to ceftazidime.

PHARMACOKINETICS
- **Absorption:** It is rapidly and almost completely absorbed on intramuscular injection and mean peak plasma concentrations about 14 and 30 micrograms/ml have been reported about 1.5 hours after doses of 500 mg and 1g respectively. Within 30 minutes of similar intravenous doses, peak plasma concentration of about 40 and 80 micrograms/ml are achieved. The plasma half-life of Cefepime is about 2 hours and is prolonged in patients with renal impairment.
- **Protein Binding:** About 20% of Cefepime is bound to plasma proteins.
- **Distribution:** Cefepime is widely distributed in body tissues and fluids. High concentrations have been detected in breast milk.
- **Elimination:** Cefepime is eliminated principally by the kidneys and about 85% of the dose is recovered unchanged in the urine. Cefepime is substantially removed by haemodialysis.

INDICATIONS
Adult
For the treatment of infections caused by susceptible bacteria:
- Lower respiratory tract infections (including pneumonia and bronchitis)
- Urinary tract infections (both complicated, including pyelonephritis and uncomplicated infections)
- Skin and skin structure infections
- Intra-abdominal infections (including peritonitis and biliary tract infections)
- Gynecologic infections
- Septicemia
- Empiric treatment in febrile neutropenia

Peadiatrics
For treatment of infections when caused by susceptible bacteria:
- Pneumonia
- Urinary tract infections (both complicated and uncomplicated, including pyelonephritis)
- Skin and skin structure infections
- Septicemia
- Empiric treatment in febrile neutropenia

Culture and susceptibility studies should be performed when appropriate to determine susceptibility of the causative organism(s) to cefepime.

Empiric therapy with hovid Sefepim for Injection may be instituted before results of susceptibility studies are known; however, once these results become available, the antibiotic treatment should be adjusted accordingly. Because of its broad spectrum of bactericidal activity against gram-positive and gram-negative bacteria, hovid Sefepim for Injection can be used as monotherapy prior to identification of the causative organism(s). In patients who are at risk of mixed aerobic-anaerobic infection, particularly if bacteria not susceptible to cefepime may be present, concurrent initial therapy with anti-anaerobic agent is recommended before the causative agents may or may not be necessary, depending on the susceptibility profile.

CONTRAINDICATIONS
This medication should not be used by patients with known allergy to cephalosporins, penicillins, penicillin derivatives or penicillamine.

PRECAUTIONS
Parenteral dosage of Cefepime should be modified in renal impairment and patients undergoing haemodialysis.

USE IN PREGNANCY AND LACTATION
This drug should be used during pregnancy only if clearly needed.Cefepime is excreted in human breast milk in very low concentrations. Caution should be used when cefepime is administered to a nursing woman.

MAIN SIDE/ADVERSE EFFECTS
Hypersensitivity reactions including skin rashes, urticaria, eosinophilia, fever, reactions resembling serum sickness, and anaphylaxis. There might be a positive response to the Coombs' Test although haemolytic anaemia rarely occours. Gastrointestinal adverse effects such as nausea, vomiting and diarrhea have been reported rarely.

Prolonged use may results in overgrowth of non-susceptible organisms and, as with other broad-spectrum antibacterials, pseudomembranous colitis may develop.

DRUG INTERACTIONS
The pharmacokinetics of cefepime and amikacin when given concurrently to patients with normal renal function do not appear to be altered.

Concomitant administration of probenecid prolongs the renal tubular secretion of cephalosporins resulting in high serum concentrations of these antibiotics.

INCOMPATIBILITIES
Cefepime should not be added to Ampicillin solutions of a strength greater than 40mg/ml or to Aminophylline, Gentamycin, Metronidazole, Netilmicin, Tobramycin or Vancomycin solutions. If these medications are administered concurrently with Cefepime, they should be administered at separate sites.

OVERDOSE
Symptom
Convulsions and other signs of CNS toxicity have been associated with high doses, especially in patients with severe renal impairment.

Treatment
Hemodialysis may be used to aid the removal of Cefepime.

DOSAGE AND ADMINISTRATION
hovid Sefepim for Injection can be administered either intravenously or intramuscularly.

The dosage and route vary according to the susceptibility of the causative organisms, the severity of the infection, and the overall condition and renal function of the patient.

Adults
Guidelines for dosage of Hovid Sefepim for Injection for adults with normal renal function are provided in Table1.

TABLE 1 - Recommended dosage schedule for Adults with Normal Renal Function (aged 12 years and older)

SITE AND TYPE OF INFECTION	DOSE	FREQUENCY
Mild to moderate urinary tract infections (uncomplicated and complicated)	500mg & 1g IV or IM	q12h
Mild to moderate infections including bronchitis, skin and skin-structure infections	1g IV or IM	q12h
Severe infections including pneumonia, urinary tract infections, complicated intraabdominal infections, including cases with an associated bacteremia	2g IV	q12h
Empiric treatment of fever in neutropenic patients	2g IV	q8h

* Usual duration of therapy is 7-10 days; more severe infections may require longer treatment. In the treatment of beta-haemolytic streptococcal infections a therapeutic dose must be administered for at least 10 days. For empirical treatment of Febrile neutropenia, usual duration of therapy is 7 days or until resolution of neutropenia.

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BACK

Paediatrics (aged 1 month up to 12 years with normal renal function)
Usual Recommended dosages:
Pneumonia, urinary tract infections, and skin structure infections:
Patients >2 months of age with body weight <40 kg: 50 mg/kg q12h for 10 days. For more severe infections, a dosage schedule of q8h can be used.

Empiric treatment of febrile neutropenia:
Patients >2 months of age with body weight <40 kg: 50 mg/kg q8h for 7-10 days.

Experience with the use of Cefepime in paediatric patients <2 months of age is limited. While this experience has been attained using the 50 mg/kg dose, modelling of pharmacokinetic data obtained in patients >2 months of age suggests that a dosage of 30 mg/kg q12h or q8h may be considered for patients aged 1 month up to 2 months.

Administration of Cefepime in these patients should be carefully monitored.

For paediatric patients with body weights >40 kg, adult dosing recommendations apply (see Table 1).

For patients older than 12 years who are <40 kg, the dosage recommendations for younger patients <40 kg should be used.

Dosage in paediatric patients should not exceed the maximum recommended dosage in adults (2g q8h).

Experience with intramuscular administration in paediatric patients is limited.

Elderly
Dose adjustment is not required, unless there is concurrent renal impairment.

Impaired hepatic function
No adjustment is necessary for patients with impaired hepatic function.

Impaired renal function
The initial dose of cefepime is the same as in patients with normal renal function.
The recommended maintenance doses of cefepime in patients with renal insufficiency are presented in Table 2.

TABLE 2 - Maintenance dosing schedule in Adult patients with renal impairment *

Creatinine clearance (mL/min)	Recommended Maintenance Dosage			
	Usual dose, no adjustment necessary			
	Very severe or life-threatening infection	Severe infection	Mild to moderate infections other than UTI	Mild to moderate urinary tract infection
> 50	2g q8h	2g q12h	1g q12h	500mg q12h
30 - 50	1g q8h	2g q24h	1g q24h	500 mg q24h
11 - 29	1g q12h	1g q24h	500 mg q24h	500 mg q24h
< 10	1g q24h	500 mg q24h	250 mg q24h	250 mg q24h

* The initial dose is the same as in patients with normal renal function.

When only a serum creatinine measurement is available, the following formula (Cockcroft and Gault equation) may be used to estimate creatinine clearance. The serum creatinine should represent a steady state of renal function:

Males : Creatinine clearance (mL/min)

$$= \frac{\text{Weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females: 0.85 x value calculated using the formula for males.

Dialysis Patients
In patients undergoing haemodialysis, approximately 68% of the total amount of cefepime present in the body at the start of dialysis will be removed during a 3 hour dialysis period.

A repeat dose, equivalent to the initial dose, should be given at the completion of each dialysis session. In patients undergoing continuous ambulatory peritoneal dialysis, cefepime may be administered at the same doses recommended for patients with normal renal function, i.e., 500 mg, 1g or 2g depending on infection severity, but at a dosage interval of every 48 hours.

Children with Impaired Renal Function
Since urinary excretion is the primary route of elimination of cefepime in paediatric patients , an adjustment of the dosage of Cefepime should also be considered in patients <12 years of age with renal impairment.

PREPARATION OF SOLUTION AND DIRECTION FOR USE
Cefepime powder is to be constituted using the volumes of diluent shown in Table 3; the diluents to be used are identified following this table.

TABLE 3 - Preparations of solutions of hovid Sefepim for Injection

	Amount of diluent to be added (mL)	Approx. available volume (mL)	Approx. cefepime concentration (mg/mL)
Intravenous 1g vial	10	11.4	90
Intra-muscular 1g vial	3	4.4	230

Intravenous (IV) administration:
The IV route of administration is preferable for patients with severe or life-threatening infections, particularly if the possibility of shock is present.

For direct IV administration, constitute Cefepime with Sterile Water for Injection, 5% Dextrose Injection or 0.9% Sodium Chloride, using the diluent volumes shown in Table 3. The resulting solution should be injected directly into the vein over a period of three to five minutes or injected into the tubing of an administration set while the patient is receiving a compatible IV fluid.

For intravenous infusion, constitute the 1g vial as noted above for direct IV administration then, add the appropriate quantity of the resulting solution to an IV container with one of the compatible IV fluids identified under Compatibility and Stability. IV infusions of a volume between 50 mL and 100 mL should be administered over a period of approximately 30 minutes.

Intramuscular (IM) administration:
Cefepime should be constituted with one of the following diluents using the volumes shown in Table 3: Sterile Water for Injection, 0.9% Sodium Chloride Injection, 5% Dextrose Injection then administered by deep IM injection into a large muscle mass (such as the upper outer quadrant of the gluteus maximus).

Although cefepime can be constituted with 0.5% or 1.0% Lidocaine hydrochloride, it is usually not necessary because cefepime causes little or no pain upon IM administration.

COMPATIBILITY AND STABILITY
Intravenous: Cefepime is compatible with one of the following IV infusion fluids : 0.9% Sodium Chloride Injection, 5% and 10% Dextrose Injection, 5% Dextroxe and 0.9% Sodium Chloride Injection, Lactate Ringers and 5% Dextrose Injection. These IV solutions retain their potency for up to 24 hours at below 22°C or for 7 days if refrigerated (2°C to 8°C).

Intramuscular: Cefepime reconstituted as directed in Table 3 will retain their potency for up to 24 hours at below 22°C or for 7 days if refrigerated (2°C to 8°C) when using the following diluents : Sterile Water for Injection, 0.9% Sodium Chloride, 5% Dextrose Injection, or 0.5% or 1% Lidocaine hydrochloride.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage: Powder-Store below 30°C. Protect from light and moisture. Reconstituted solution-Refer to compatibility and stability.

Dosage Forms / Packaging Available:
The vials are packed in boxes of 1, 5, 10, 25 and 50.

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